

Chatbots tend to operate in one of two ways—either via machine learning or with set guidelines.

Rasa

Rasa helps in creating virtual assistants, which are used to automate human-to-computer interactions across platforms ranging from websites to social media. It supplies conversational AI infrastructure for a global community of developers, providing the tools to build chat-based and voice-based contextual assistants.

Rasa is powered by open source software and runs across organisations ranging from startups to Fortune 500s, and industries like healthcare, financial services, retail and insurance. It has three main functions, which together provide everything needed to build a virtual assistant.

Natural language understanding: It converts raw text from user messages into structured data, parses the user's intent and extracts important key details.

Dialogue management: Machine learning-powered dialogue management decides what the assistant should do next, based on the user's message and context from the conversation.

Integrations: It has built-in integration points for over 10 messaging channels, plus endpoints to connect with databases, APIs, and other data sources.

Installation

The first step is to install Rasa.

```
% python -m venv env
% source env/bin/activate
% pip install rasa

For UI: pip install rasa-x -i https://pypi.rasa.com/
simple [optional]

% rasa init # setup the basic file structure
% update config.yml and endpoint.yml [optional]
% rasa train # train your model
    % rasa train nlu
    % rasa train core

% rasa shell # run the bot from shell
% rasa run actions # action server
% rasa data validate # validate if the data given in the
configuration is good
% rasa run -m models --enable-api --cors "*" --debug # run

the rasa as rest http server
```

A sample UI installation (not associated with Rasa) is given here. To have conversations with Rasa in the user interface (UI), there is a module called chatroom. It is ReactJS based (URL: <https://github.com/scalableminds/chatroom>).

```
% mkdir ui; cd ui
% git clone https://github.com/scalableminds/chatroom.git
% cd chatroom
% yarn install
% yarn build (after any customised changes)
% yarn serve # to run the ui server
% update the index.html with the address of the rasa server
running
```

Open the URL <http://127.0.0.1:8080/index.html> in Chrome, as seen in Figure 1. (Note: Chrome is the supported browser for now.)

Figure 2 shows the sample conversation, which was written from scratch using Rasa.

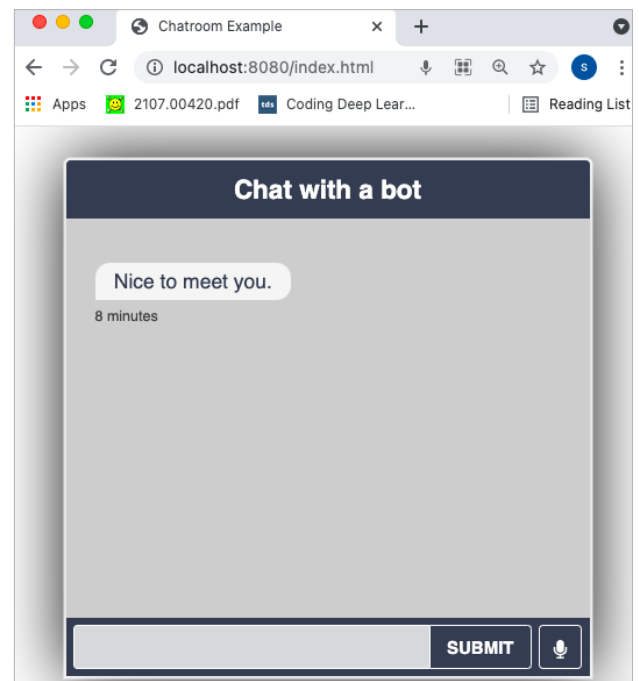


Figure 1: Initial page loaded using the chatroom module

```
Bot loaded. Type a message and press enter (use '/stop' to exit):
Your input -> hi
Hey! How can I help you?
Your input -> who are you
I am a bot, powered by Rasa.
Your input -> may I know when college reops
The college is not open now we are still working through online
Your input -> my rollnumber is 1234567890
I'll remember your rollnumber 1234567890!
Your input -> can you help me with my results
All done!
I am going to run a result search using the following parameters:
- rollnumber: 1234567890
For 1234567890, result is Pass with 98 score.
Your input -> may I know the fees structure
For 1234567890, fees is 10000 INR.
Your input -> I have moved from civil
How was your experience with the department.
Your input -> will be moving to computer
Wish you best luck in the new department.
Your input -> /stop
```

Figure 2: Sample Rasa conversation using Rasa shell